Retail Customer Segmentation for Targeted Marketing

Business Analytics Portfolio Case Study

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Executive view

This case study segments 3,000 loyalty-card customers over a six-month period into five actionable customer groups. The analysis converts transactional and category-level data into customer-level behavioural features, corrects a major bakery-spend data issue, applies scaling and PCA, and uses K-Means clustering to support targeted marketing decisions.

3,000	31 inputs	13 (~80%)	5	0.171
Customers analysed	Feature set	PCA components	Final clusters	Silhouette

1. Business context and objective

The retailer needed a practical segmentation of its loyalty customer base to guide marketing planning. The project used six months of point-of-sale data to describe customers by how often they shop, how much they spend, how broad their baskets are, how recently they visited, and which product categories they buy.

The objective was not to produce a predictive sales model, but to create interpretable customer archetypes that marketing teams can use to prioritise retention, reactivation and basket-building campaigns.

2. Data preparation and feature engineering

The analysis combined four transactional sources: customer summary data, basket-level visits, category-level spend and line-item purchases. The modelling table was built at one row per customer.

Stage	
Data cleaning	Currency-style fields were converted to numeric values. Negative spends were treated as data-quality corrections and set to zero rather than deleting customers from the sample.
Bakery correction	Bakery spend had been recorded as zero in the category-spend table. It was recalculated from line-item data, changing 2,963 of 3,000 customers and lifting mean bakery spend from £0.00 to £38.21.
Feature blocks	Features captured frequency, recency, total/average spend, basket size, basket breadth and category preferences.
Redundancy control	Obvious duplicate summary fields were removed. A correlation screen then removed tenure_days and active_days because they overlapped strongly with recency_days and n_baskets.
Scaling and PCA	All 31 retained inputs were standardised before PCA and clustering. The final clustering used 13 principal components, retaining approximately 80% of the behavioural signal.

3. Segmentation methodology

K-Means was selected because it is efficient, transparent and appropriate for producing similarity-based customer groups from standardised behavioural features. Solutions from k=4 to k=10 were compared using silhouette scores and business interpretability. Although k=4 had the highest numerical silhouette score, the project brief required 5-7 segments and k=5 produced usable segment sizes with clearer business narratives than higher-k alternatives.

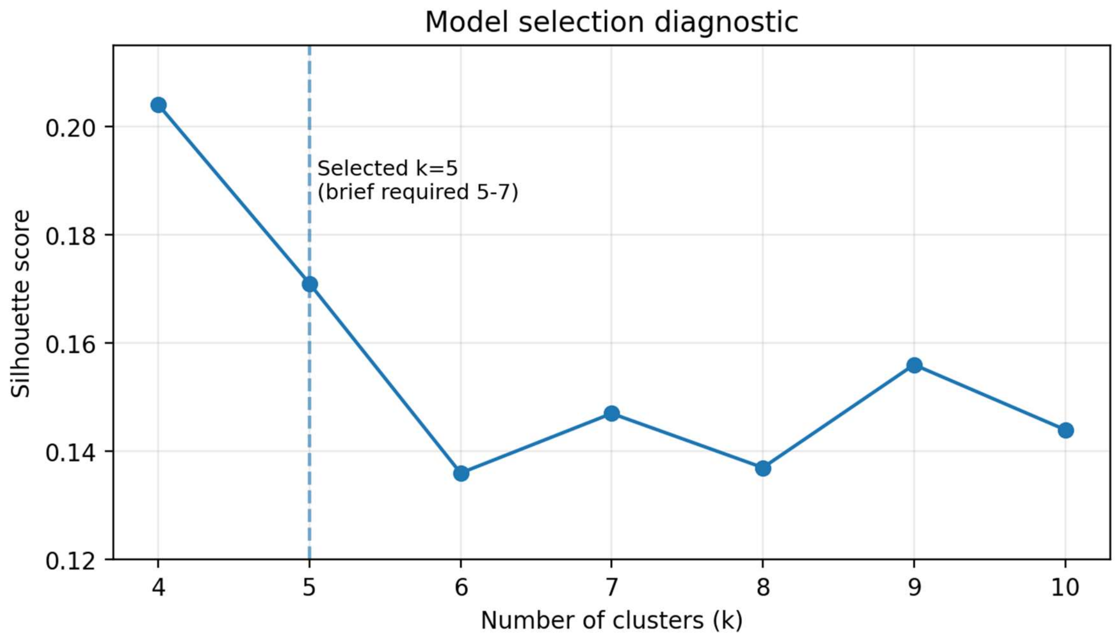

Figure 1. Silhouette diagnostic used to support the selected five-segment solution.

4. Segment overview

The five-cluster solution reveals a strong value and mission gradient. One large mainstream group accounts for 45.0% of customers, while smaller groups show distinctive shopping missions such as premium multi-category trips, very frequent top-ups and periodic stock-up behaviour.

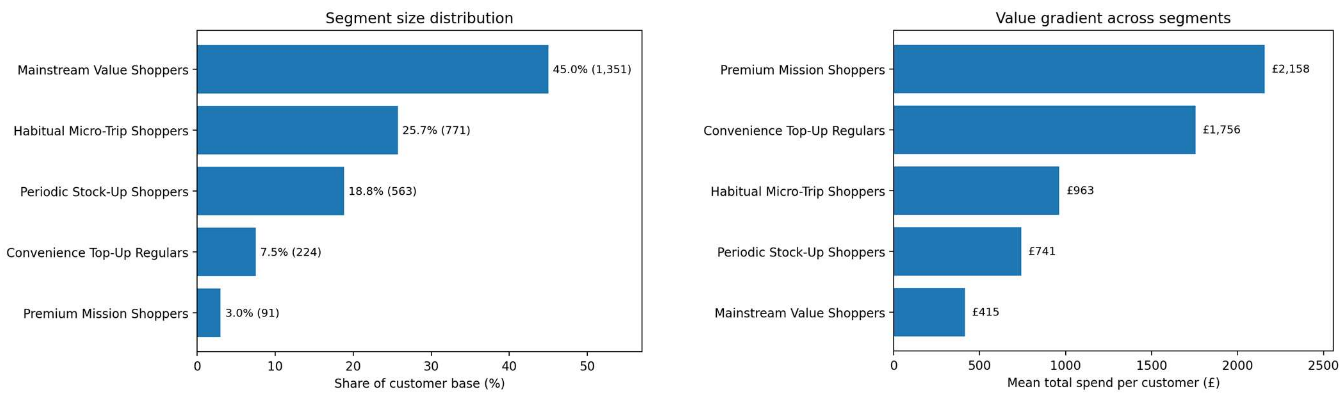

Figure 2. Segment scale and mean total-spend gradient.

Table 1 summarises the core commercial KPIs behind each segment.

Segment	Share	Customers	Visits	Recency	Avg basket	Total spend
Mainstream Value Shoppers	45.0%	1,351	45.3	11.8d	£10.63	£415
Premium Mission Shoppers	3.0%	91	61.8	3.5d	£44.79	£2,158
Convenience Top-Up Regulars	7.5%	224	126.8	1.9d	£15.78	£1,756
Habitual Micro-Trip Shoppers	25.7%	771	106.5	1.9d	£10.04	£963
Periodic Stock-Up Shoppers	18.8%	563	32.3	11.0d	£26.11	£741

Segment	Categories/trip	Primary interpretation
Mainstream Value Shoppers	4.00	Large base; lower spend and lower freshness. Best served through low-cost mass targeting.
Premium Mission Shoppers	7.63	Small but very high value; large, broad baskets. Priority retention segment.
Convenience Top-Up Regulars	5.25	Very frequent and recent; strong tobacco/ancillary signal. Protect habit and availability.
Habitual Micro-Trip Shoppers	4.04	High footfall with low basket value. Best opportunity is add-on basket

		building.
Periodic Stock-Up Shoppers	6.56	Fewer visits but larger baskets. Reactivation and restock timing are key levers.

5. Pen portraits

Segment	Business pen portrait
Mainstream Value Shoppers	The quiet majority: lower value, less recent and broad enough for scalable generic offers. Campaigns should be low-cost because the return per customer is limited.
Premium Mission Shoppers	Small but commercially important: high total spend, high average basket spend, large quantities and broad missions. The main goal is retention and share-of-wallet protection.
Convenience Top-Up Regulars	Highly frequent, highly recent customers with elevated tobacco, lottery and cashpoint behaviour. The risk is habit disruption through queues, stock-outs or poor convenience.
Habitual Micro-Trip Shoppers	Large, frequent, low-basket segment. They already visit often, so marketing should focus on adding one or two relevant items per trip rather than increasing visit frequency.
Periodic Stock-Up Shoppers	Less frequent but higher basket-value shoppers with broad category coverage. They are suitable for timed restock reminders, bundle offers and reactivation triggers.

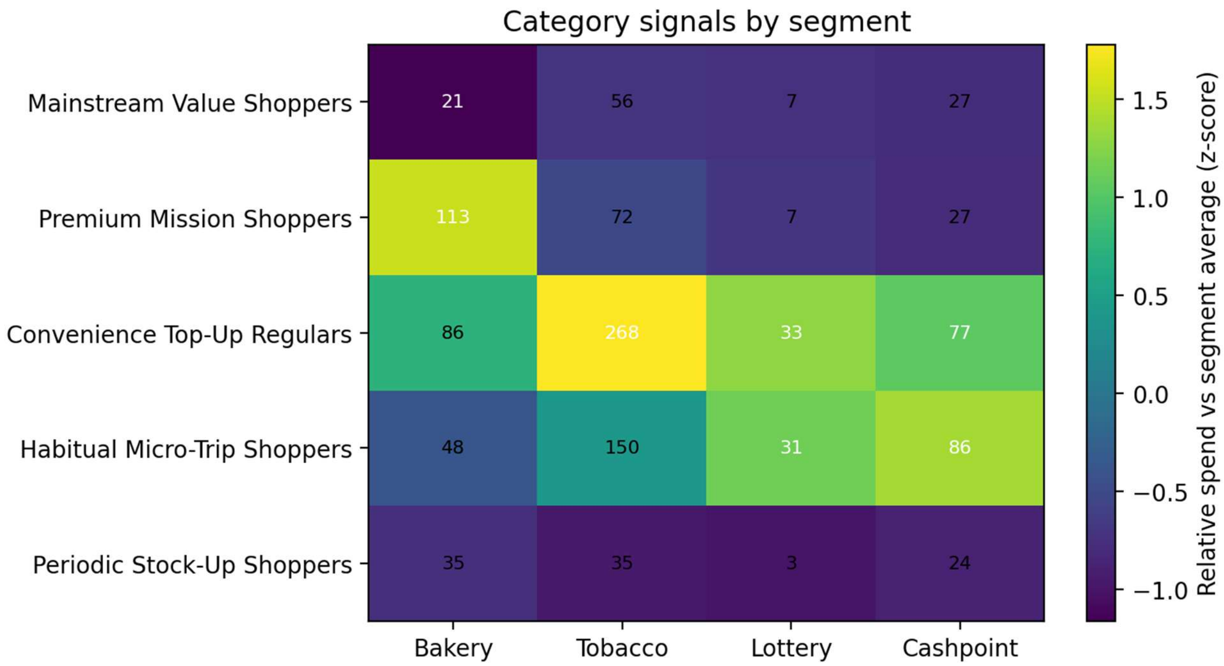

Figure 3. Category signals supporting the segment pen portraits. Cell labels show mean category spend (£).

6. Recommended marketing priorities

For a portfolio audience, the strongest recommendation is to treat the segmentation as a decision-support layer for campaign testing. The recommended targets balance customer value, reachability and likely marketing levers.

Target	Role	Recommended action
Priority 1: Premium Mission Shoppers	Retention and value protection	Use VIP-style benefits, personalised multi-category bundles and service/availability messages. This group is only 3.0% of customers but has the highest mean total spend (£2,158) and the highest average basket spend (£44.79).
Priority 2: Periodic Stock-Up Shoppers	Reactivation and basket building	Use timed restock prompts, cross-category bundles and moderate multi-buy offers. They buy larger baskets (£26.11 average basket spend) but are less recent on average (11.03 days), creating a clear timing opportunity.
Secondary opportunity: Habitual Micro-Trip Shoppers	Incremental basket expansion	Use impulse add-ons, meal-deal style offers and checkout/app prompts. They are already highly engaged, so the commercial opportunity is trip monetisation rather than visit generation.

7. Limitations and next steps

Area	Portfolio-grade interpretation
Cluster quality	The silhouette score is modest (0.171), which is common in overlapping retail behaviour. Segments should be validated through campaign response, not treated as perfect natural groups.
Margin and promotion response	The dataset captures spend and category behaviour but not product margin, campaign cost or promotion response. These should be added before using the segments for budget allocation.
Customer identity and time horizon	The analysis uses a six-month sample. Segment stability should be monitored over time and customers should be refreshed periodically as their behaviour changes.
Recommended next analysis	Run A/B tests by segment, build uplift models for offer response, and connect segmentation to churn/retention monitoring.

Appendix: technical traceability

Item	Detail
Customers	3,000 loyalty customers observed over six months.
Inputs after cleaning	31 features retained after duplicate-feature and high-correlation removal.
PCA	13 components used for clustering, retaining approximately 80% of variance; 18 components would be needed for approximately 90%.
Final model	K-Means, k=5, n_init=20, random_state=42, fitted on 13-component PCA representation.
Segment mapping	Customer clutter 1-5 mapped to recruiter-facing segment names in this report.

Data availability note: raw transactional files are not included in the public portfolio because they were supplied for academic use. The public repository should include the cleaned report, methodology notebook and a README explaining that raw data is excluded.